EXCHANGE-ONLY QUANTUM CONTROL AND STABILIZER EXTRACTION

A verified numerical research synthesis (R138-R154)

Numerical working paper. Author list and publication status not yet confirmed.
This PDF is a text-focused repository rendition. For figures and typeset equations consult the original ChatGPT PDF archive.

Scope and evidence standard

This paper is a working synthesis of independently packaged numerical research rounds R138-R154, not a claim of newly completed hardware experiments or an accepted publication. It reorganizes the evidence into a single scientific narrative, and deliberately retains negative results. The included archives contain the original optimized angles, physical exchange schedules, numerical certificates, simulation code, and model assumptions. Numerical endpoints reported as "exact" mean residuals at approximately 10^{-11} or better in the specified floating-point physical representations, not symbolic proofs of the angles.

The guiding question is whether simultaneous exchange offers a physically efficient path from isolated, noise-shaped EO entanglers to encoded stabilizer measurements with acceptable syndrome error, leakage, and duration. Three distinctions are essential: physical vs. encoded correction, gate vs. measurement-instrument equivalence, and conditional numerical fidelity vs. hardware error rates.

Exchange-only encoding and physical controls

Each logical EO qubit is carried by three spin-$1/2$ particles. In the total-spin $S=1/2$ manifold, the state of a chosen pair (singlet or triplet) represents a logical degree of freedom, while $S_z=\pm 1/2$ gives a gauge degree of freedom; the $S=3/2$ quartet is leakage. For two triples there are $4\times4=16$ encoded logical--gauge input states within the full six-spin dimension $2^6=64$.

Writing the physical singlet projector on spins i,j as

$$
P_{ij}^{(s)}=\tfrac12(I-\mathrm{SWAP}_{ij}),
$$

our dimensionless positive exchange convention is

$$
X_{ij}(a)=\exp(i\pi a P_{ij}^{(s)}),\qquad 0.15\leq a<2
$$

for active controls in the reported searches. Unlike a sequential pulse decomposition, a simultaneous layer on overlapping bonds must be evaluated with one matrix exponential:

$$
U_\ell=\exp\!\left(i\pi\sum_{b\in\mathcal B_\ell}
a_{\ell b}P_b^{(s)}\right),\quad U=U_L\cdots U_1.
$$

Nonoverlapping exchanges commute; overlapping exchanges usually do not. Every target must be checked on all encoded gauge states, not on a selected logical basis state alone.

If E is the 64×16 code isometry, an exact encoded CZ with fixed gauge has $UE=E(CZ\otimes I_g)$, with vanishing component in the orthogonal leakage space. For measurement protocols we also admit a consistent gauge-only transformation and record-dependent overall phase when the resulting quantum instrument, not the full unmeasured unitary, is unchanged.

Noise model and acceptance metrics

The main model samples a fractional static gain δ_b once per physical bond per shot and reuses that value at every occurrence of the bond. A distinct independent fluctuation $\eta_{\ell b}$ perturbs each active control. The representative assumptions are

$$
\delta_b\sim\mathcal N(0,0.015^2),\qquad
\eta_{\ell b}\sim\mathcal N(0,0.0006^2).
$$

These are the assumptions used by the accompanying numerical studies, not a site-resolved calibration of a particular experimental device. Separate context-dependent coactivation fluctuations were tested with assumed standard deviations and various correlation times; those values should not be read as experimental measurements.

The input-referred first-order static generator on bond b is

$$
K_b=-iU(0)^\dagger\left.\frac{\partial U}{\partial\delta_b}\right|_0.
$$

We report its gauge-subtracted harmful coefficient, including leakage,

$$
H_b=\frac1{16}\min_{h=h^\dagger}
\left\|K_b-E(I_{\rm logical}\otimes h_{\rm gauge})\right\|_F^2,
\qquad H_{\rm pb}=\sum_bH_b.
$$

$H_{\rm IID}$ similarly sums the independent-control contributions. For sufficiently weak fluctuations, infidelity begins as a quadratic form in the variances. Gate-level $H_{\rm pb}$, conditional instrument loss, wrong-syndrome probability, and leakage probability are different observables with different normalizations and must not be interchanged. "Controls" counts active physical exchange couplings; "ASAP depth" assumes unit-duration compatible layers and is not a calibrated wall-clock time.

Six-spin simultaneous-exchange CZ: the central constructive result

The short exact CZ reference is the 22-control Fong--Wandzura-derived sequence in the archived linear six-spin encoding [1]. Later rounds explored exact simultaneous replacement, error-generator cones, local spin-sector recurrences, positive-exchange composite loops, and six-spin noncommuting frames.

For the directly comparable 28-control/19-layer R139 baseline, the optimized R143 sequence uses 30 active controls in the same 19 nominal ASAP layers and gives

$$
H_{\rm pb}:37.668119847\longrightarrow32.011918153
\qquad(15.02\%\ \mathrm{reduction}),
$$

while $H_{\rm IID}$ decreases from 45.7993 to 43.5393. The physical gate is validated with full 64-dimensional exchange propagation and, after translation, with 512-dimensional propagation of a nine-spin assembly. The encoded residual is on the order of 10^{-13}. A second 30-control trade-off has $H_{\rm pb}=32.9022$, $H_{34}=8.4960$, and $H_{\rm IID}=41.9486$. Neither eliminates the bridge error at first order.

Figure available in the companion PDF and reproducibility archive.

Certified CZ reference	Active controls	ASAP depth	$H_{\rm pb}$	H_{34}	$H_{\rm IID}$
R139 total-noise baseline	28	19	37.6681	8.8616	45.7993
R143 optimized static	30	19	32.0119	8.9451	43.5393
R143 bridge-balanced	30	19	32.9022	8.4960	41.9486

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R144 selective bridge | 32 | 20 | 49.7808 | 7.8607 | 42.2649
R149 six-spin coactivation | 35 | 19 | 37.9950 | 8.9451 | 51.3420
```

The improvement is principally a change in J_{12} logical-error sensitivity, not a complete bridge-gain correction. Specifically, in the R143 comparison, J_{12} harmful contribution falls from approximately 12.2520 to 7.4315, while J_{34} changes from 8.8616 to 8.9451. Some additional endpoint Jacobian directions have very small singular values ($\sim10^{-6}$), limiting the physical significance of a simple rank-count argument.

Figure available in the companion PDF and reproducibility archive.

Why straightforward compensation failed

Several carefully bounded negative screens were useful. Adding exact simultaneous identity loops, including four-spin revival blocks, generally worsened $H_{\rm pb}$ because each additional positive exchange accumulates static and IID error. Some optimizations reduce H_{34} at the cost of substantially increasing other bond sensitivities. Introducing J_{56} as an extra control in the old five-spin frame creates new endpoint directions, but their positive-area tangent cone cannot be cancelled by arbitrarily small added pulses near that particular seed. This is not a no-go theorem against all six-spin CZ designs.

The six-spin nested commutator

$$
D=[P_{56}^{(s)},[P_{45}^{(s)},P_{34}^{(s)}]]
$$

is nonzero even though $[P_{56}^{(s)},P_{34}^{(s)}]=0$, and a two-bond direction $Q=i[P_{45}^{(s)},P_{34}^{(s)}]$ gives greater reduction of the projected error in idealized operator-budget screens. However, the budget coefficients do not correspond to physically executable pulse areas. Later rounds found positive-exchange simultaneous loops with either sign of a Q component, but none gave a lower total-noise exact CZ after insertion. Positive exchange yields semidefinite first-order physical generators, so a pure, unaccompanied traceless Q term is not physically available as a bond-gain derivative.

A three-spin dual-static-gain corrected exchange: useful but not transferable

On a crossing triple $2,3,4$, a six-layer, twelve-control positive simultaneous sequence was synthesized that implements a physical $J_{34}(1)$ SWAP. With two independent bond-static gains,

$$
U_\ell=\exp\{i\pi[a_\ell P_{23}^{(s)}+b_\ell P_{34}^{(s)}]\},
$$

and the first-order generators reduce, numerically, to scalars on the triple's $S=1/2$ sector:

$$
K_{23}=\kappa_{23}\Pi_{234}^{S=1/2},\qquad
K_{34}=\kappa_{34}\Pi_{234}^{S=1/2}.
$$

Consequently, the isolated three-spin logical operation has no first-order harmful static-gain rotation. A deterministic noise-amplitude scan gives logical infidelity power-law slopes close to four on both independent error directions, versus approximately two for a direct SWAP. The 15,000-shot isolated-triple simulations yield 3.72×10^{-4} versus 9.88×10^{-6} for direct versus composite SWAP under the static-plus-IID model.

Figure available in the companion PDF and reproducibility archive.

The protection does not carry over to two EO qubits. The projector $\Pi_{234}^{S=1/2}$ is not gauge-only for the $123|456$ code. Replacing one bridge pulse in an otherwise exact 22-control CZ creates a 31-control exact gate with $H_{\rm pb}=80.276$ and $H_{34}=11.584$, both worse than the original. If simultaneous context noise fluctuates independently every layer rather than being shot-static, the isolated triple's fourth-order benefit also degrades markedly. These observations support noise correlation-time-aware compilation rather than a simple claim of universal exchange-gain immunity.

Measurement-instrument-first design

For a Pauli check S and recorded outcome $m\in\{0,1\}$, the target ideal nondemolition Kraus operator on data is

$$
M_m=P_m^{S}=\frac{I+(-1)^mS}{2}.
$$

Syndrome probabilities alone are insufficient. A valid measurement must preserve arbitrary superpositions within each eigenspace of S and must not introduce unwanted logical backaction or leakage. For example, separately measuring Z_1Z_2 and Z_3Z_4 then multiplying the recorded bits coarse-grains two stronger measurements; it is not the same instrument as measuring $Z_1Z_2Z_3Z_4$. On $|++++\rangle$, conditioning on positive four-body parity gives purity 1 for the correct projector but only $1/2$ for the fine-grained protocol.

Nine-spin coherent ZZ and XX checks

Take three consecutive EO triples, data-left $(1,2,3)$, ancilla $(4,5,6)$, and data-right $(7,8,9)$. The exact baseline is

$$
H_A\;CZ_{L,A}\;CZ_{A,R}\;H_A\;\mathrm{ST\ readout}_A,
$$

where each logical H_A is synthesized with positive exchange only, not with an assumed physical single-spin magnetic rotation. The 50-control/29-layer ZZ construction and its 62-control XX counterpart are validated by complete physical 512-dimensional propagation and outcome-conditioned Kraus comparison across all 32 input logical--gauge states.

Exact measurement-specific elimination of two controls

An important reduction is exact for the recorded instrument but not necessarily for the full unmeasured unitary. At preparation the ancilla is known to be $|0,g\rangle$, with its first pair $(4,5)$ singlet. Thus

$$
X_{45}(a)|0,g\rangle=e^{i\pi a}|0,g\rangle.
$$

At readout the projectors are $M_0=P_{45}^{(s)}$ and $M_1=I-P_{45}^{(s)}$. As $X_{45}(a)$ commutes with both,

$$
M_0X_{45}(a)=e^{i\pi a}M_0,\qquad
M_1X_{45}(a)=M_1.
$$

The initial and final J_{45} controls can therefore be omitted without changing the normalized outcome-conditioned physical data operation (only irrelevant outcome phases remain). This takes the ZZ instrument from 50 controls / 29 ASAP layers to 48 controls / 27 layers. This exact pruning alone does not improve the error statistics, because the discarded controls were physically irrelevant even when miscalibrated.

The additional gain comes from choosing a different legal two-exchange ancilla preparation branch. In chronological order, the optimized preparation uses $J_{56}(0.60817344797)$ followed by $J_{45}(1.69591327602)$, while the readout basis changer retains the archived branch. These operations prepare the same required ancilla input but need not implement the same H on arbitrary ancillary states. Full nine-spin two-outcome Kraus residuals are $\sim10^{-15}$.

```
Nine-spin ZZ instrument | Controls | ASAP depth | Static instrument curvature | Static wrong-syndrome curvature
R153 original | 50 | 29 | 90.1754 | 50.1050
R154 boundary-pruned only | 48 | 27 | 90.1754 | 50.1050
R154 optimized preparation | 48 | 27 | 86.0781 | 46.0078
R153 two R143 CZ gates | 66 | 39 | 76.5499 | 46.1944
```

The optimized 48-control choice reduces wrong-syndrome curvature by 8.18% against the original 50-control instrument, while keeping the average leakage curvature approximately unchanged. A principal sensitivity is the ancilla-adjacent J_{56} bond, whose wrong-syndrome coefficient changes from approximately 20.0460 to 14.2895. Thus an instrument-level objective can prioritize a different bond than an isolated CZ compiler.

Finite-noise assessment

With 750 paired simulations at shot-static gain 1.5% and independent jitter 0.06%, the original 50-control instrument has wrong-syndrome probability 0.01103592, and the optimized 48-control instrument has 0.01004484. The paired difference is -0.00099108 ± 0.00026945 (quoted 95% normal confidence-interval halfwidth). Instrument loss changes from 0.01974105 to 0.01873975. Leakage differences are statistically inconclusive. These figures do not include calibrated readout errors, waiting dephasing or reset.

Figure available in the companion PDF and reproducibility archive.

The 66-control check has lower full-instrument loss and leakage under another matched comparison but requires 18 extra controls and 12 extra ASAP layers relative to 48-control. Its wrong-syndrome advantage is not statistically established in that comparison. In particular, one should not merge sample means from separate Monte Carlo batches into a single paired statistical test.

Fully physical weight-four reference on an explicit star graph

A five-EO reference uses an ancilla triple $(1,2,3)$ and data triples $(4,5,6)$, $(7,8,9)$, $(10,11,12)$, $(13,14,15)$. It assumes physical controllable exchange edges $(3,4)$, $(3,7)$, $(3,10)$, and $(3,13)$ plus specified internal bonds. This is not a strict one-dimensional nearest-neighbor device and must not be described as one.

Four mapped 22-control FW-derived CZs plus a two-control ancilla preparation and a two-control readout changer yield a 92-control / 56-ASAP-layer coherent ZZZZ measurement baseline. Applying physical three-exchange logical Hadamards to all four data triples before and after the ZZZZ sequence yields a 116-control / 56-layer XXXX baseline in the stated unit-time concurrency model.

All 16 data logical patterns multiplied by all 32 combined gauge patterns, i.e. 512 encoded input basis states, were explicitly propagated in the full $2^{15}=32768$ physical state space. Maximum output vector residuals are approximately 2.45×10^{-15} for ZZZZ and 3.31×10^{-15} for XXXX; selected coherent-superposition checks verify preservation within each parity sector. These are CZ-factorized constructions. They do not demonstrate a shorter direct four-body measurement, repeated rounds of correction, or a practical star-chip implementation.

Interpretation, limitations and open research questions

The trajectory supports three defensible conclusions. First, overlapping simultaneous exchanges provide real noncommuting control and allow nontrivial noise-generator engineering, but a short six-spin, all-bond first-order-corrected exact CZ remains an open problem in these archives. Second, a strong locally corrected crossing triple is not necessarily a corrected multi-EO gate: the code and leakage projectors must be re-evaluated globally. Third, the most tangible system-level saving comes from relaxing unnecessary intermediate gate constraints when ancilla initialization and measurement are fixed.

The major limitations are: (i) noise strengths and context covariance are model inputs rather than measured per-device maps; (ii) ASAP depth is not a waveform-calibrated duration and 48 controls do not necessarily imply a smaller total exchange-area time; (iii) no independent noise calibration of the claimed star couplers is provided; (iv) singlet-triplet readout does not separate logical triplet from $S=3/2$ leakage; (v) no repeated syndrome, ancilla reset, hook-error circuit, leakage-reduction unit, decoder, logical error suppression or threshold was evaluated; and (vi) this synthesis is not a full novelty or experimental feasibility review.

The next useful publication-quality advance would be a directly synthesized quantum instrument with fewer than 48 controls or a materially lower wrong-syndrome/error-plus-duration objective, validated in full nine-spin dynamics. For weight-four checks, the analogous objective must be formulated on an explicitly feasible geometry, include readout/reset/leakage and repeated-cycle propagation, and compare full conditional Kraus maps rather than parity bits alone.

Reproducibility and research provenance

The accompanying repository package retains selected full-precision sequences and four source-archive ZIPs (R143, R146, R153, R154); an additional all-round archival bundle contains R138-R154. Each source ZIP is treated as a frozen research artifact with its own code, evidence and internal SHA-256 listing. Re-run the corresponding archive verifiers before changing geometry or gauge convention; note that their denominators and definitions differ between gate, isolated triple, and instrument metrics. Exact filenames, command entry points, and hashes are listed in REPRODUCIBILITY.md and SOURCE_MANIFEST.csv.

References

[1] B. H. Fong and S. M. Wandzura, "Universal quantum computation and leakage reduction in the 3-qubit decoherence free subsystem," Quantum Information and Computation 11, 1003-1018 (2011). https://doi.org/10.26421/QIC11.11-12-9

[2] G. T. Hickman, X. Wang, J. P. Kestner, and S. Das Sarma, "Dynamically corrected gates for an exchange-only qubit," Physical Review B 88, 161303(R) (2013). https://doi.org/10.1103/PhysRevB.88.161303

[3] I. Heinz et al., "Fast Quantum Gates for Exchange-Only Qubits Using Simultaneous Exchange Pulses," PRX Quantum 6, 030353 (2025). https://doi.org/10.1103/njq3-fcdd

[4] M. T. Madzik et al., "Operating two exchange-only qubits in parallel," Nature 647, 870-875 (2025). https://doi.org/10.1038/s41586-025-09767-5

[5] Members of the HRL Quantum Team and Collaborators, "A digitally controlled silicon quantum processing unit," Nature 655, 1154-1159 (2026). https://doi.org/10.1038/s41586-026-10754-7

[6] The numerical research archives R138-R154 accompanying this synthesis (October 2026), including reproducibility scripts, pulse CSVs, full-state validation outputs, and model-specific Monte Carlo evidence. These are internal computational source records, not independently peer-reviewed publications.